ESTUDIANTES:

VICTOR JOSE DE PEÑA BERNARD

WALDY STEVEN GOMEZ MONTERO

MATRICULAS:

2023-0506

2025-0783

FACILITADOR:

LISBETT JAQUEZ MUESES

TEMA:

FUNDAMENTOS DE REDES

PROYECTO FINAL

FECHA:

22/04/2026

Índice

Presentación

Este proyecto tiene como objetivo el diseño e implementación de una red utilizando direccionamiento IPv6, aplicando los conocimientos adquiridos sobre configuración de routers, asignación de direcciones y conectividad entre diferentes segmentos de red.

Se desarrolló una topología de red que interconecta múltiples LANs a través de routers, permitiendo la comunicación eficiente entre distintas sucursales. Para ello, se configuraron servicios esenciales como DHCPv6 para la asignación dinámica de direcciones IP, así como acceso remoto seguro mediante SSH.

Además, se implementó un protocolo de enrutamiento que garantiza la correcta propagación de rutas entre los dispositivos, asegurando la conectividad total de la red. También se aplicaron configuraciones básicas de seguridad y se realizaron pruebas de conectividad para validar el correcto funcionamiento de toda la infraestructura.

Este proyecto integra conceptos clave de redes modernas, destacando el uso de IPv6 como base fundamental para el direccionamiento y la escalabilidad de la red.

2. Requerimientos presentados

1. Diseñar el direccionamiento IPv6 utilizando la dirección **2003:BEBA:XXXX::/48**, donde *XXXX* representa los últimos 4 dígitos de la matrícula.
2. Configurar la red, realizando las configuraciones necesarias en los routers correspondientes.
3. Configurar acceso remoto seguro en los routers mediante **SSH**, utilizando:

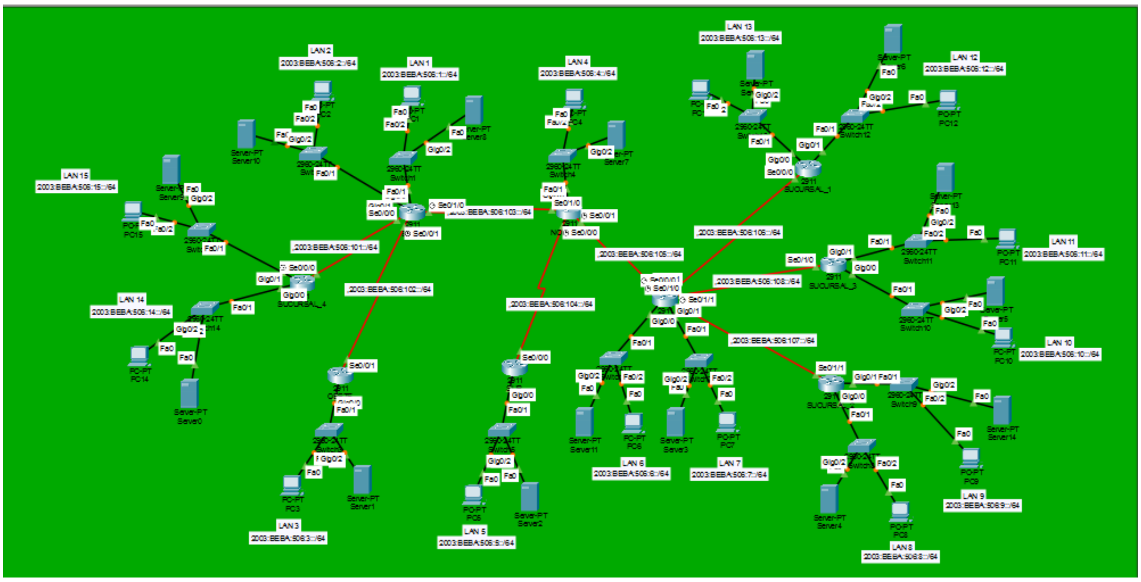
    - Usuario: *Manager*
    - Clave: *Class01*

4. Utilizar un protocolo de enrutamiento para garantizar la comunicación entre redes.
5. Implementar un servidor **DHCPv6 en cada LAN** para la asignación dinámica de direcciones a los hosts, utilizando:

    - DNS: **2002:FEA:CAC0::2/64**
    - Dominio: **globals.com**

6. Aplicar configuraciones básicas de seguridad en los dispositivos de red.
7. Verificar la conectividad entre los diferentes segmentos de la red.
8. Documentar la red incluyendo: tabla de direccionamiento, topología y configuraciones principales.

3. Topología lógica trabajada



4. Tabla de direccionamiento

LAN	Router	Interfaz	Dirección Global (GUA)	Link-Local
LAN 5	SUR	G0/0	2003:BEBA:506:5::1/64	FE80::5:1
LAN 6	SEDE	G0/0	2003:BEBA:506:6::1/64	FE80::6:1
LAN 7	SEDE	G0/1	2003:BEBA:506:7::1/64	FE80::7:1
LAN 8	SUCURSAL_2	G0/0	2003:BEBA:506:8::1/64	FE80::8:1
LAN 9	SUCURSAL_2	G0/1	2003:BEBA:506:9::1/64	FE80::9:1

LAN	Router	Interfaz	Dirección Global (GUA)	Link-Local
LAN 10	SUCURSAL_3	G0/0	2003:BEBA:506:10::1/64	FE80::10:1
LAN 11	SUCURSAL_3	G0/1	2003:BEBA:506:11::1/64	FE80::11:1
LAN 12	SUCURSAL_1	G0/1	2003:BEBA:506:12::1/64	FE80::12:1
LAN 13	SUCURSAL_1	G0/0	2003:BEBA:506:13::1/64	FE80::13:1
LAN 14	SUCURSAL_4	G0/0	2003:BEBA:506:14::1/64	FE80::14:1
LAN 15	SUCURSAL_4	G0/1	2003:BEBA:506:15::1/64	FE80::15:1

5. Configuracion de equipos de red (Routers)

ROUTER ESTE

Show Running-config

!

version 15.1

no service timestamps log datetime msec

no service timestamps debug datetime msec

service password-encryption

!

hostname ESTE

!

```
!
!
!
!
!
!
!
no ip cef
ipv6 unicast-routing
!
no ipv6 cef
!
!
!
username Manager secret 5 $1$mERr$9xcyzgMubYEUu5jrby8ds.
!
!
license udi pid CISCO2911/K9 sn FTX1524K2C4-
!
!
!
!
!
!
!
!
!
!
```

```
!
ip ssh version 2
no ip domain-lookup
ip domain-name red.local
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0
 description LAN 1
 no ip address
 duplex auto
 speed auto
 ipv6 address FE80::1:1 link-local
 ipv6 address 2003:BEBA:506:1::1/64
 ipv6 nd other-config-flag
 ipv6 nd managed-config-flag
 ipv6 dhcp server LAN_1
!
interface GigabitEthernet0/1
 description LAN 2
 no ip address
 duplex auto
```

```
 speed auto
 ipv6 address FE80::2:1 link-local
 ipv6 address 2003:BEBA:506:2::1/64
 ipv6 nd other-config-flag
 ipv6 nd managed-config-flag
 ipv6 dhcp server LAN_2
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Serial0/0/0
 description Conexion con SUCURSAL-4
 no ip address
 ipv6 address FE80::101:2 link-local
 ipv6 address 2003:BEBA:506:101::2/64
 ipv6 rip ITLA_RIP enable
!
interface Serial0/0/1
 description Conexion con OESTE
 no ip address
 ipv6 address FE80::102:1 link-local
 ipv6 address 2003:BEBA:506:102::1/64
 ipv6 rip ITLA_RIP enable
 clock rate 2000000
!
```

```
interface Serial0/1/0
 description Conexin con NORESTE
 no ip address
 ipv6 address FE80::103:1 link-local
 ipv6 address 2003:BEBA:506:103::1/64
 ipv6 rip ITLA_RIP enable
 clock rate 2000000
!
interface Serial0/1/1
 no ip address
 ipv6 rip ITLA_RIP enable
 clock rate 2000000
 shutdown
!
interface Serial0/2/0
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/2/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/3/0
 no ip address
 clock rate 2000000
 shutdown
```

```
!
interface Serial0/3/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
ipv6 router rip ITLA_RIP
 redistribute connected
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
 exec-timeout 5 0
 password 7 0822455D0A165546
 login
!
```

line aux 0

!

line vty 0 4

 login local

 transport input ssh

!

!

!

end

Show IPv6 Route

ESTE#Show IPv6 Route

IPv6 Routing Table - 29 entries

Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP

 U - Per-user Static route, M - MIPv6

 I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary

 ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

 O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

 ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2

 D - EIGRP, EX - EIGRP external

C 2003:BEBA:506:1::/64 [0/0]

 via GigabitEthernet0/0, directly connected

L 2003:BEBA:506:1::1/128 [0/0]

 via GigabitEthernet0/0, receive

C 2003:BEBA:506:2::/64 [0/0]

 via GigabitEthernet0/1, directly connected

L 2003:BEBA:506:2::1/128 [0/0]

 via GigabitEthernet0/1, receive

R 2003:BEBA:506:3::/64 [120/2]

via FE80::102:2, Serial0/0/1

R 2003:BEBA:506:4::/64 [120/16]

via FE80::103:2, Serial0/1/0

R 2003:BEBA:506:5::/64 [120/16]

via FE80::103:2, Serial0/1/0

R 2003:BEBA:506:6::/64 [120/16]

via FE80::103:2, Serial0/1/0

R 2003:BEBA:506:7::/64 [120/16]

via FE80::103:2, Serial0/1/0

R 2003:BEBA:506:8::/64 [120/16]

via FE80::103:2, Serial0/1/0

R 2003:BEBA:506:9::/64 [120/16]

via FE80::103:2, Serial0/1/0

R 2003:BEBA:506:10::/64 [120/16]

via FE80::103:2, Serial0/1/0

R 2003:BEBA:506:11::/64 [120/16]

via FE80::103:2, Serial0/1/0

R 2003:BEBA:506:12::/64 [120/16]

via FE80::103:2, Serial0/1/0

R 2003:BEBA:506:13::/64 [120/16]

via FE80::103:2, Serial0/1/0

R 2003:BEBA:506:14::/64 [120/2]

via FE80::101:1, Serial0/0/0

R 2003:BEBA:506:15::/64 [120/2]

via FE80::101:1, Serial0/0/0

C 2003:BEBA:506:101::/64 [0/0]

via Serial0/0/0, directly connected

L 2003:BEBA:506:101::2/128 [0/0]

via Serial0/0/0, receive

C 2003:BEBA:506:102::/64 [0/0]

via Serial0/0/1, directly connected

L 2003:BEBA:506:102::1/128 [0/0]

via Serial0/0/1, receive

C 2003:BEBA:506:103::/64 [0/0]

via Serial0/1/0, directly connected

L 2003:BEBA:506:103::1/128 [0/0]

via Serial0/1/0, receive

R 2003:BEBA:506:104::/64 [120/16]

via FE80::103:2, Serial0/1/0

R 2003:BEBA:506:105::/64 [120/16]

via FE80::103:2, Serial0/1/0

R 2003:BEBA:506:106::/64 [120/16]

via FE80::103:2, Serial0/1/0

R 2003:BEBA:506:107::/64 [120/16]

via FE80::103:2, Serial0/1/0

R 2003:BEBA:506:108::/64 [120/16]

via FE80::103:2, Serial0/1/0

L FF00::/8 [0/0]

via Null0, receiveL FF00::/8 [0/0]

Show IPv6 interface brief

```
GigabitEthernet0/0          [up/up]
    FE80::1:1
    2003:BEBA:506:1::1
GigabitEthernet0/1          [up/up]
    FE80::2:1
    2003:BEBA:506:2::1
GigabitEthernet0/2          [administratively down/down]
    unassigned
Serial0/0/0                 [up/up]
    FE80::101:2
    2003:BEBA:506:101::2
Serial0/0/1                 [up/up]
    FE80::102:1
    2003:BEBA:506:102::1
Serial0/1/0                 [up/up]
    FE80::103:1
    2003:BEBA:506:103::1
Serial0/1/1                 [administratively down/down]
    unassigned
Serial0/2/0                 [administratively down/down]
    unassigned
Serial0/2/1                 [administratively down/down]
    unassigned
Serial0/3/0                 [administratively down/down]
    unassigned
Serial0/3/1                 [administratively down/down]
    unassigned
Vlan1                       [administratively down/down]
    unassigned
```

NORESTE

Show Running-config

!

version 15.1

no service timestamps log datetime msec

no service timestamps debug datetime msec

service password-encryption

!

hostname NORESTE

!

!

!

!

!

!

!

!

no ip cef

```
ipv6 unicast-routing
!
no ipv6 cef
!
!
!
username Manager secret 5 $1$mERr$9xcyzgMubYEUu5jrby8ds.
!
!
license udi pid CISCO2911/K9 sn FTX15244881-
!
!
!
!
!
!
!
!
!
!
!
ip ssh version 2
no ip domain-lookup
ip domain-name red.local
!
!
spanning-tree mode pvst
!
!
!
!
```

```
!
!
interface GigabitEthernet0/0
 description LAN 4
 no ip address
 duplex auto
 speed auto
 ipv6 address FE80::4:1 link-local
 ipv6 address 2003:BEBA:506:4::1/64
 ipv6 nd other-config-flag
 ipv6 nd managed-config-flag
 ipv6 dhcp server LAN_4
!
interface GigabitEthernet0/1
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Serial0/0/0
 description Hacia SUR
 no ip address
 ipv6 address FE80::104:1 link-local
 ipv6 address 2003:BEBA:506:104::1/64
 ipv6 rip ITLA_RIP enable
```

```
 clock rate 2000000
!
interface Serial0/0/1
 description Hacia SEDE
 no ip address
 ipv6 address FE80::105:1 link-local
 ipv6 address 2003:BEBA:506:105::1/64
 ipv6 rip ITLA_RIP enable
 clock rate 64000
!
interface Serial0/1/0
 description Hacia ESTE
 no ip address
 ipv6 address FE80::103:2 link-local
 ipv6 address 2003:BEBA:506:103::2/64
 ipv6 rip ITLA_RIP enable
!
interface Serial0/1/1
 no ip address
 ipv6 rip ITLA_RIP enable
 clock rate 2000000
 shutdown
!
interface Serial0/2/0
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/2/1
 no ip address
 clock rate 2000000
```

```
 shutdown
!
interface Serial0/3/0
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/3/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
ipv6 router rip ITLA_RIP
 redistribute connected
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
 exec-timeout 5 0
```

```
 password 7 0822455D0A165546
 login
!
line aux 0
!
line vty 0 4
 login local
 transport input ssh
!
!
end
```

Show IPv6 Route

```
NORESTE#Show IPv6 Route
IPv6 Routing Table - 17 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
R   2003:BEBA:506:1::/64 [120/1]
     via FE80::103:1, Serial0/1/0
R   2003:BEBA:506:2::/64 [120/1]
     via FE80::103:1, Serial0/1/0
R   2003:BEBA:506:3::/64 [120/3]
     via FE80::103:1, Serial0/1/0
C   2003:BEBA:506:4::/64 [0/0]
```

via GigabitEthernet0/0, directly connected

L 2003:BEBA:506:4::1/128 [0/0]

via GigabitEthernet0/0, receive

R 2003:BEBA:506:5::/64 [120/2]

via FE80::104:2, Serial0/0/0

R 2003:BEBA:506:14::/64 [120/3]

via FE80::103:1, Serial0/1/0

R 2003:BEBA:506:15::/64 [120/3]

via FE80::103:1, Serial0/1/0

R 2003:BEBA:506:101::/64 [120/2]

via FE80::103:1, Serial0/1/0

R 2003:BEBA:506:102::/64 [120/2]

via FE80::103:1, Serial0/1/0

C 2003:BEBA:506:103::/64 [0/0]

via Serial0/1/0, directly connected

L 2003:BEBA:506:103::2/128 [0/0]

via Serial0/1/0, receive

C 2003:BEBA:506:104::/64 [0/0]

via Serial0/0/0, directly connected

L 2003:BEBA:506:104::1/128 [0/0]

via Serial0/0/0, receive

C 2003:BEBA:506:105::/64 [0/0]

via Serial0/0/1, directly connected

L 2003:BEBA:506:105::1/128 [0/0]

via Serial0/0/1, receive

L FF00::/8 [0/0]

via Null0, receive

Show IPv6 interface brief

```
GigabitEthernet0/0          [up/up]
    FE80::4:1
    2003:BEBA:506:4::1
GigabitEthernet0/1          [administratively down/down]
    unassigned
GigabitEthernet0/2          [administratively down/down]
    unassigned
Serial0/0/0                 [up/up]
    FE80::104:1
    2003:BEBA:506:104::1
Serial0/0/1                 [up/up]
    FE80::105:1
    2003:BEBA:506:105::1
Serial0/1/0                 [up/up]
    FE80::103:2
    2003:BEBA:506:103::2
Serial0/1/1                 [administratively down/down]
    unassigned
Serial0/2/0                 [administratively down/down]
    unassigned
Serial0/2/1                 [administratively down/down]
    unassigned
Serial0/3/0                 [administratively down/down]
    unassigned
Serial0/3/1                 [administratively down/down]
    unassigned
Vlan1                       [administratively down/down]
```

OESTE

Show Running-config

!

version 15.1

no service timestamps log datetime msec

no service timestamps debug datetime msec

service password-encryption

!

hostname OESTE

!

!

!

!

!

!

!

!

no ip cef

```
ipv6 unicast-routing
!
no ipv6 cef
!
ipv6 dhcp pool LAN_3
 address prefix 2003:BEBA:506:3::/64 lifetime 172800 86400
 dns-server 2002:FEA:CAC0::2
 domain-name globals.com
!
!
!
username Manager secret 5 $1$mERr$9xcyzgMubYEUu5jrby8ds.
!
!
license udi pid CISCO2911/K9 sn FTX1524RGB5-
!
!
!
!
!
!
!
!
!
!
!
ip ssh version 2
no ip domain-lookup
```

```
ip domain-name red.local
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0
 description LAN 3
 no ip address
 duplex auto
 speed auto
 ipv6 address FE80::3:1 link-local
 ipv6 address 2003:BEBA:506:3::1/64
 ipv6 nd other-config-flag
 ipv6 nd managed-config-flag
 ipv6 rip ITLA_RIP enable
 ipv6 dhcp server LAN_3
!
interface GigabitEthernet0/1
 no ip address
 duplex auto
 speed auto
 shutdown
!
```

```
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Serial0/0/0
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/0/1
 description Conexion con ESTE
 no ip address
 ipv6 address FE80::102:2 link-local
 ipv6 address 2003:BEBA:506:102::2/64
 ipv6 rip ITLA_RIP enable
!
interface Serial0/1/0
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/1/1
 no ip address
 clock rate 2000000
 shutdown
!
```

```
interface Serial0/2/0
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/2/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/3/0
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/3/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
ipv6 router rip ITLA_RIP
!
ip classless
!
```

```
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
 exec-timeout 5 0
 password 7 0822455D0A165546
 login
!
line aux 0
!
line vty 0 4
 login local
 transport input ssh
!
!
!
end
```

Show IPv6 Route

IPv6 Routing Table - 15 entries

Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP

U - Per-user Static route, M - MIPv6

I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2

D - EIGRP, EX - EIGRP external

R 2003:BEBA:506:1::/64 [120/1]

via FE80::102:1, Serial0/0/1

R 2003:BEBA:506:2::/64 [120/1]

via FE80::102:1, Serial0/0/1

C 2003:BEBA:506:3::/64 [0/0]

via GigabitEthernet0/0, directly connected

L 2003:BEBA:506:3::1/128 [0/0]

via GigabitEthernet0/0, receive

R 2003:BEBA:506:4::/64 [120/2]

via FE80::102:1, Serial0/0/1

R 2003:BEBA:506:5::/64 [120/4]

via FE80::102:1, Serial0/0/1

R 2003:BEBA:506:14::/64 [120/3]

via FE80::102:1, Serial0/0/1

R 2003:BEBA:506:15::/64 [120/3]

via FE80::102:1, Serial0/0/1

R 2003:BEBA:506:101::/64 [120/2]

via FE80::102:1, Serial0/0/1

C 2003:BEBA:506:102::/64 [0/0]

via Serial0/0/1, directly connected

L 2003:BEBA:506:102::2/128 [0/0]

via Serial0/0/1, receive

R 2003:BEBA:506:103::/64 [120/2]

via FE80::102:1, Serial0/0/1

R 2003:BEBA:506:104::/64 [120/3]

via FE80::102:1, Serial0/0/1

R 2003:BEBA:506:105::/64 [120/3]

via FE80::102:1, Serial0/0/1

L FF00::/8 [0/0]

via Null0, receive

Show IPv6 interface brief

```
GigabitEthernet0/0          [up/up]
    FE80::3:1
    2003:BEBA:506:3::1
GigabitEthernet0/1          [administratively down/down]
    unassigned
GigabitEthernet0/2          [administratively down/down]
    unassigned
Serial0/0/0                 [administratively down/down]
    unassigned
Serial0/0/1                 [up/up]
    FE80::102:2
    2003:BEBA:506:102::2
Serial0/1/0                 [administratively down/down]
    unassigned
Serial0/1/1                 [administratively down/down]
    unassigned
Serial0/2/0                 [administratively down/down]
    unassigned
Serial0/2/1                 [administratively down/down]
    unassigned
Serial0/3/0                 [administratively down/down]
    unassigned
Serial0/3/1                 [administratively down/down]
    unassigned
Vlan1                       [administratively down/down]
    unassigned
```

SEDE

Show Running-config

!

version 15.1

no service timestamps log datetime msec

no service timestamps debug datetime msec

```
service password-encryption
!
hostname SEDE
!
!
!
!
!
!
!
!
ip cef
ipv6 unicast-routing
!
no ipv6 cef
!
!
!
username Manager privilege 15 secret 5 $1$mERr$9xcyzgMubYEUu5jrby8ds.
!
!
license udi pid CISCO2911/K9 sn FTX1524G1ZZ-
!
!
!
!
!
!
!
!
!
```

```
!
!
ip ssh version 2
ip domain-name red.local
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0
 description LAN
 no ip address
 duplex auto
 speed auto
 ipv6 address FE80::6:1 link-local
 ipv6 address 2003:BEBA:506:6::1/64
 ipv6 nd other-config-flag
 ipv6 nd managed-config-flag
 ipv6 dhcp server LAN_6
!
interface GigabitEthernet0/1
 description LAN
 no ip address
 duplex auto
 speed auto
 ipv6 address FE80::7:1 link-local
 ipv6 address 2003:BEBA:506:7::1/64
```

```
 ipv6 nd other-config-flag
 ipv6 nd managed-config-flag
 ipv6 dhcp server LAN_7
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Serial0/0/0
 description Conexin con SUCURSAL_1
 no ip address
 ipv6 address FE80::106:1 link-local
 ipv6 address 2003:BEBA:506:106::1/64
 ipv6 rip ITLA_RIP enable
 clock rate 2000000
!
interface Serial0/0/1
 description Hacia NORESTE
 no ip address
 ipv6 address FE80::105:2 link-local
 ipv6 address 2003:BEBA:506:105::2/64
 ipv6 rip ITLA_RIP enable
!
interface Serial0/1/0
 description Hacia SUCURSAL_3
 no ip address
 ipv6 address FE80::108:1 link-local
 ipv6 address 2003:BEBA:506:108::1/64
 ipv6 rip ITLA_RIP enable
```

```
 clock rate 2000000
!
interface Serial0/1/1
 description Hacia SUCURSAL_2
 no ip address
 ipv6 address FE80::107:1 link-local
 ipv6 address 2003:BEBA:506:107::1/64
 ipv6 rip ITLA_RIP enable
 clock rate 2000000
!
interface Serial0/2/0
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/2/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/3/0
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/3/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Vlan1
```

```
 no ip address
 shutdown
!
ipv6 router rip ITLA_RIP
 redistribute connected
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
 password 7 0822455D0A165546
 login
!
line aux 0
!
line vty 0 4
 login local
 transport input ssh
!
!
!
end
```

Show IPv6 Route

IPv6 Routing Table - 19 entries

Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP

U - Per-user Static route, M - MIPv6

I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2

D - EIGRP, EX - EIGRP external

C 2003:BEBA:506:6::/64 [0/0]

via GigabitEthernet0/0, directly connected

L 2003:BEBA:506:6::1/128 [0/0]

via GigabitEthernet0/0, receive

C 2003:BEBA:506:7::/64 [0/0]

via GigabitEthernet0/1, directly connected

L 2003:BEBA:506:7::1/128 [0/0]

via GigabitEthernet0/1, receive

R 2003:BEBA:506:8::/64 [120/2]

via FE80::107:2, Serial0/1/1

R 2003:BEBA:506:9::/64 [120/2]

via FE80::107:2, Serial0/1/1

R 2003:BEBA:506:10::/64 [120/2]

via FE80::108:2, Serial0/1/0

R 2003:BEBA:506:11::/64 [120/2]

via FE80::108:2, Serial0/1/0

R 2003:BEBA:506:12::/64 [120/2]

via FE80::106:2, Serial0/0/0

R 2003:BEBA:506:13::/64 [120/2]

via FE80::106:2, Serial0/0/0

C 2003:BEBA:506:105::/64 [0/0]

via Serial0/0/1, directly connected

L 2003:BEBA:506:105::2/128 [0/0]

via Serial0/0/1, receive

C 2003:BEBA:506:106::/64 [0/0]

via Serial0/0/0, directly connected

L 2003:BEBA:506:106::1/128 [0/0]

via Serial0/0/0, receive

C 2003:BEBA:506:107::/64 [0/0]

via Serial0/1/1, directly connected

L 2003:BEBA:506:107::1/128 [0/0]

via Serial0/1/1, receive

C 2003:BEBA:506:108::/64 [0/0]

via Serial0/1/0, directly connected

L 2003:BEBA:506:108::1/128 [0/0]

via Serial0/1/0, receive

L FF00::/8 [0/0]

via Null0, receive

Show IPv6 interface brief

```
GigabitEthernet0/0          [up/up]
    FE80::6:1
    2003:BEBA:506:6::1
GigabitEthernet0/1          [up/up]
    FE80::7:1
    2003:BEBA:506:7::1
GigabitEthernet0/2          [administratively down/down]
    unassigned
Serial0/0/0                 [up/up]
    FE80::106:1
    2003:BEBA:506:106::1
Serial0/0/1                 [up/up]
    FE80::105:2
    2003:BEBA:506:105::2
Serial0/1/0                 [up/up]
    FE80::108:1
    2003:BEBA:506:108::1
Serial0/1/1                 [up/up]
    FE80::107:1
    2003:BEBA:506:107::1
Serial0/2/0                 [administratively down/down]
    unassigned
Serial0/2/1                 [administratively down/down]
    unassigned
Serial0/3/0                 [administratively down/down]
    unassigned
Serial0/3/1                 [administratively down/down]
    unassigned
Vlan1                       [administratively down/down]
    unassigned
```

<u>SUR</u>

<u>Show Running-config</u>

!

version 15.1

no service timestamps log datetime msec

no service timestamps debug datetime msec

service password-encryption

!

hostname SUR

!

!

```
!
!
!
!
!
!
no ip cef
ipv6 unicast-routing
!
no ipv6 cef
!
!
!
username Manager secret 5 $1$mERr$9xcyzgMubYEUu5jrby8ds.
!
!
license udi pid CISCO2911/K9 sn FTX15241K3J-
!
!
!
!
!
!
!
!
!
!
!
```

```
ip ssh version 2
no ip domain-lookup
ip domain-name red.local
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0
 description LAN 5
 no ip address
 duplex auto
 speed auto
 ipv6 address FE80::5:1 link-local
 ipv6 address 2003:BEBA:506:5::1/64
 ipv6 nd other-config-flag
 ipv6 nd managed-config-flag
 ipv6 rip ITLA_RIP enable
 ipv6 dhcp server LAN_5
!
interface GigabitEthernet0/1
 no ip address
 duplex auto
 speed auto
```

```
 shutdown
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Serial0/0/0
 description Conexin con NOROESTE
 no ip address
 ipv6 address FE80::104:2 link-local
 ipv6 address 2003:BEBA:506:104::2/64
 ipv6 rip ITLA_RIP enable
!
interface Serial0/0/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/1/0
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/1/1
 no ip address
 clock rate 2000000
```

```
 shutdown
!
interface Serial0/2/0
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/2/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/3/0
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/3/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
ipv6 router rip ITLA_RIP
!
```

```
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
 exec-timeout 5 0
 password 7 0822455D0A165546
 login
!
line aux 0
!
line vty 0 4
 login local
 transport input ssh
!
!
!
end
```

Show IPv6 Route

IPv6 Routing Table - 19 entries

Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP

U - Per-user Static route, M - MIPv6

I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2

D - EIGRP, EX - EIGRP external

C 2003:BEBA:506:6::/64 [0/0]

via GigabitEthernet0/0, directly connected

L 2003:BEBA:506:6::1/128 [0/0]

via GigabitEthernet0/0, receive

C 2003:BEBA:506:7::/64 [0/0]

via GigabitEthernet0/1, directly connected

L 2003:BEBA:506:7::1/128 [0/0]

via GigabitEthernet0/1, receive

R 2003:BEBA:506:8::/64 [120/2]

via FE80::107:2, Serial0/1/1

R 2003:BEBA:506:9::/64 [120/2]

via FE80::107:2, Serial0/1/1

R 2003:BEBA:506:10::/64 [120/2]

via FE80::108:2, Serial0/1/0

R 2003:BEBA:506:11::/64 [120/2]

via FE80::108:2, Serial0/1/0

R 2003:BEBA:506:12::/64 [120/2]

via FE80::106:2, Serial0/0/0

R 2003:BEBA:506:13::/64 [120/2]

via FE80::106:2, Serial0/0/0

C 2003:BEBA:506:105::/64 [0/0]

via Serial0/0/1, directly connected

L 2003:BEBA:506:105::2/128 [0/0]

via Serial0/0/1, receive

C 2003:BEBA:506:106::/64 [0/0]

via Serial0/0/0, directly connected

L 2003:BEBA:506:106::1/128 [0/0]

via Serial0/0/0, receive

C 2003:BEBA:506:107::/64 [0/0]

via Serial0/1/1, directly connected

L 2003:BEBA:506:107::1/128 [0/0]

via Serial0/1/1, receive

C 2003:BEBA:506:108::/64 [0/0]

via Serial0/1/0, directly connected

L 2003:BEBA:506:108::1/128 [0/0]

via Serial0/1/0, receive

L FF00::/8 [0/0]

via Null0, receive

Show IPv6 interface brief

```
GigabitEthernet0/0           [up/up]
    FE80::6:1
    2003:BEBA:506:6::1
GigabitEthernet0/1           [up/up]
    FE80::7:1
    2003:BEBA:506:7::1
GigabitEthernet0/2           [administratively down/down]
    unassigned
Serial0/0/0                  [up/up]
    FE80::106:1
    2003:BEBA:506:106::1
Serial0/0/1                  [up/up]
    FE80::105:2
    2003:BEBA:506:105::2
Serial0/1/0                  [up/up]
    FE80::108:1
    2003:BEBA:506:108::1
Serial0/1/1                  [up/up]
    FE80::107:1
    2003:BEBA:506:107::1
Serial0/2/0                  [administratively down/down]
    unassigned
Serial0/2/1                  [administratively down/down]
    unassigned
Serial0/3/0                  [administratively down/down]
    unassigned
Serial0/3/1                  [administratively down/down]
    unassigned
Vlan1                        [administratively down/down]
    unassigned
```

SUCURSAL 1

!

version 15.1

no service timestamps log datetime msec

no service timestamps debug datetime msec

service password-encryption

!

hostname SUCURSAL_1

!

!

!

!

!

!

!

!

no ip cef

```
ipv6 unicast-routing
!
no ipv6 cef
!
!
!
username Manager secret 5 $1$mERr$9xcyzgMubYEUu5jrby8ds.
!
!
license udi pid CISCO2911/K9 sn FTX1524G9JX-
!
!
!
!
!
!
!
!
!
!
!
ip ssh version 2
no ip domain-lookup
ip domain-name red.local
!
!
spanning-tree mode pvst
!
!
!
!
```

```
!
!
interface GigabitEthernet0/0
 description LAN_SUC1_H1
 no ip address
 duplex auto
 speed auto
 ipv6 address FE80::13:1 link-local
 ipv6 address 2003:BEBA:506:13::1/64
 ipv6 nd other-config-flag
 ipv6 nd managed-config-flag
 ipv6 rip ITLA_RIP enable
 ipv6 dhcp server LAN_13
!
interface GigabitEthernet0/1
 description LAN_SUC1_H2
 no ip address
 duplex auto
 speed auto
 ipv6 address FE80::12:1 link-local
 ipv6 address 2003:BEBA:506:12::1/64
 ipv6 nd other-config-flag
 ipv6 nd managed-config-flag
 ipv6 rip ITLA_RIP enable
 ipv6 dhcp server LAN_12
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
```

```
!
interface Serial0/0/0
 description Conexin con SEDE
 no ip address
 ipv6 address FE80::106:2 link-local
 ipv6 address 2003:BEBA:506:106::2/64
 ipv6 rip ITLA_RIP enable
!
interface Serial0/0/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/1/0
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/1/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/2/0
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/2/1
 no ip address
 clock rate 2000000
```

```
 shutdown
!
interface Serial0/3/0
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/3/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
ipv6 router rip ITLA_RIP
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
 exec-timeout 5 0
 password 7 0822455D0A165546
```

login

!

line aux 0

!

line vty 0 4

login local

transport input ssh

!

!

!

end

Show IPv6 Route

IPv6 Routing Table - 16 entries

Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP

U - Per-user Static route, M - MIPv6

I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2

D - EIGRP, EX - EIGRP external

R 2003:BEBA:506:6::/64 [120/1]

via FE80::106:1, Serial0/0/0

R 2003:BEBA:506:7::/64 [120/1]

via FE80::106:1, Serial0/0/0

R 2003:BEBA:506:8::/64 [120/3]

via FE80::106:1, Serial0/0/0

R 2003:BEBA:506:9::/64 [120/3]

via FE80::106:1, Serial0/0/0

R 2003:BEBA:506:10::/64 [120/3]

via FE80::106:1, Serial0/0/0

R 2003:BEBA:506:11::/64 [120/3]

via FE80::106:1, Serial0/0/0

C 2003:BEBA:506:12::/64 [0/0]

via GigabitEthernet0/1, directly connected

L 2003:BEBA:506:12::1/128 [0/0]

via GigabitEthernet0/1, receive

C 2003:BEBA:506:13::/64 [0/0]

via GigabitEthernet0/0, directly connected

L 2003:BEBA:506:13::1/128 [0/0]

via GigabitEthernet0/0, receive

R 2003:BEBA:506:105::/64 [120/2]

via FE80::106:1, Serial0/0/0

C 2003:BEBA:506:106::/64 [0/0]

via Serial0/0/0, directly connected

L 2003:BEBA:506:106::2/128 [0/0]

via Serial0/0/0, receive

R 2003:BEBA:506:107::/64 [120/2]

via FE80::106:1, Serial0/0/0

R 2003:BEBA:506:108::/64 [120/2]

via FE80::106:1, Serial0/0/0

L FF00::/8 [0/0]

via Null0, receive

Show IPv6 interface brief

```
GigabitEthernet0/0          [up/up]
    FE80::13:1
    2003:BEBA:506:13::1
GigabitEthernet0/1          [up/up]
    FE80::12:1
    2003:BEBA:506:12::1
GigabitEthernet0/2          [administratively down/down]
    unassigned
Serial0/0/0                 [up/up]
    FE80::106:2
    2003:BEBA:506:106::2
Serial0/0/1                 [administratively down/down]
    unassigned
Serial0/1/0                 [administratively down/down]
    unassigned
Serial0/1/1                 [administratively down/down]
    unassigned
Serial0/2/0                 [administratively down/down]
    unassigned
Serial0/2/1                 [administratively down/down]
    unassigned
Serial0/3/0                 [administratively down/down]
    unassigned
Serial0/3/1                 [administratively down/down]
    unassigned
Vlan1                       [administratively down/down]
    unassigned
```

SUCURSAL 2

Show Running-config

!

version 15.1

no service timestamps log datetime msec

no service timestamps debug datetime msec

service password-encryption

!

hostname SUCURSAL_2

!

!

!

!

!

!

!

!

no ip cef

```
ipv6 unicast-routing
!
no ipv6 cef
!
!
!
username Manager secret 5 $1$mERr$9xcyzgMubYEUu5jrby8ds.
!
!
license udi pid CISCO2911/K9 sn FTX15246M1G-
!
!
!
!
!
!
!
!
!
!
!
ip ssh version 2
no ip domain-lookup
ip domain-name red.local
!
!
spanning-tree mode pvst
!
!
!
!
```

```
!
!
interface GigabitEthernet0/0
 description LAN
 no ip address
 duplex auto
 speed auto
 ipv6 address FE80::8:1 link-local
 ipv6 address 2003:BEBA:506:8::1/64
 ipv6 nd other-config-flag
 ipv6 nd managed-config-flag
 ipv6 rip ITLA_RIP enable
 ipv6 dhcp server LAN_8
!
interface GigabitEthernet0/1
 description LAN
 no ip address
 duplex auto
 speed auto
 ipv6 address FE80::9:1 link-local
 ipv6 address 2003:BEBA:506:9::1/64
 ipv6 nd other-config-flag
 ipv6 nd managed-config-flag
 ipv6 rip ITLA_RIP enable
 ipv6 dhcp server LAN_9
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
```

```
!
interface Serial0/0/0
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/0/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/1/0
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/1/1
 no ip address
 ipv6 address FE80::107:2 link-local
 ipv6 address 2003:BEBA:506:107::2/64
 ipv6 rip ITLA_RIP enable
!
interface Serial0/2/0
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/2/1
 no ip address
 clock rate 2000000
 shutdown
```

```
!
interface Serial0/3/0
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/3/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
ipv6 router rip ITLA_RIP
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
 exec-timeout 5 0
 password 7 0822455D0A165546
 login
```

```
!
line aux 0
!
line vty 0 4
 login local
 transport input ssh
!
!
!
end
```

Show IPv6 Route

```
IPv6 Routing Table - 16 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
U - Per-user Static route, M - MIPv6
I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
D - EIGRP, EX - EIGRP external
R 2003:BEBA:506:6::/64 [120/1]
via FE80::107:1, Serial0/1/1
R 2003:BEBA:506:7::/64 [120/1]
via FE80::107:1, Serial0/1/1
C 2003:BEBA:506:8::/64 [0/0]
via GigabitEthernet0/0, directly connected
L 2003:BEBA:506:8::1/128 [0/0]
via GigabitEthernet0/0, receive
C 2003:BEBA:506:9::/64 [0/0]
via GigabitEthernet0/1, directly connected
```

L 2003:BEBA:506:9::1/128 [0/0]

via GigabitEthernet0/1, receive

R 2003:BEBA:506:10::/64 [120/3]

via FE80::107:1, Serial0/1/1

R 2003:BEBA:506:11::/64 [120/3]

via FE80::107:1, Serial0/1/1

R 2003:BEBA:506:12::/64 [120/3]

via FE80::107:1, Serial0/1/1

R 2003:BEBA:506:13::/64 [120/3]

via FE80::107:1, Serial0/1/1

R 2003:BEBA:506:105::/64 [120/2]

via FE80::107:1, Serial0/1/1

R 2003:BEBA:506:106::/64 [120/2]

via FE80::107:1, Serial0/1/1

C 2003:BEBA:506:107::/64 [0/0]

via Serial0/1/1, directly connected

L 2003:BEBA:506:107::2/128 [0/0]

via Serial0/1/1, receive

R 2003:BEBA:506:108::/64 [120/2]

via FE80::107:1, Serial0/1/1

L FF00::/8 [0/0]

via Null0, receive

<u>Show IPv6 interface brief</u>

```
GigabitEthernet0/0          [up/up]
    FE80::8:1
    2003:BEBA:506:8::1
GigabitEthernet0/1          [up/up]
    FE80::9:1
    2003:BEBA:506:9::1
GigabitEthernet0/2          [administratively down/down]
    unassigned
Serial0/0/0                 [administratively down/down]
    unassigned
Serial0/0/1                 [administratively down/down]
    unassigned
Serial0/1/0                 [administratively down/down]
    unassigned
Serial0/1/1                 [up/up]
    FE80::107:2
    2003:BEBA:506:107::2
Serial0/2/0                 [administratively down/down]
    unassigned
Serial0/2/1                 [administratively down/down]
    unassigned
Serial0/3/0                 [administratively down/down]
    unassigned
Serial0/3/1                 [administratively down/down]
    unassigned
Vlan1                       [administratively down/down]
    unassigned
```

SUCURSAL 3

Show Running-config

!

version 15.1

no service timestamps log datetime msec

no service timestamps debug datetime msec

service password-encryption

!

hostname SUCURSAL_3

!

!

!

!

!

!

!

```
!
no ip cef
ipv6 unicast-routing
!
no ipv6 cef
!
ipv6 dhcp pool POOL_LAN
 dns-server 2002:FEA:CAC0::2
!
!
!
username Manager secret 5 $1$mERr$9xcyzgMubYEUu5jrby8ds.
!
!
license udi pid CISCO2911/K9 sn FTX1524I43Q-
!
!
!
!
!
!
!
!
!
!
!
ip ssh version 2
no ip domain-lookup
```

```
ip domain-name red.local
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0
 description LAN 10
 no ip address
 duplex auto
 speed auto
 ipv6 address FE80::10:1 link-local
 ipv6 address 2003:BEBA:506:10::1/64
 ipv6 nd other-config-flag
 ipv6 nd managed-config-flag
 ipv6 rip ITLA_RIP enable
 ipv6 dhcp server LAN_11
!
interface GigabitEthernet0/1
 description Conexin con LAN 11
 no ip address
 duplex auto
 speed auto
 ipv6 address FE80::11:1 link-local
```

```
 ipv6 address 2003:BEBA:506:11::1/64
 ipv6 nd other-config-flag
 ipv6 nd managed-config-flag
 ipv6 rip ITLA_RIP enable
 ipv6 dhcp server POOL_LAN
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Serial0/0/0
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/0/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/1/0
 no ip address
 ipv6 address FE80::108:2 link-local
 ipv6 address 2003:BEBA:506:108::2/64
 ipv6 rip ITLA_RIP enable
!
```

```
interface Serial0/1/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/2/0
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/2/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/3/0
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/3/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Vlan1
 no ip address
 shutdown
```

```
!
ipv6 router rip ITLA_RIP
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
 exec-timeout 5 0
 password 7 0822455D0A165546
 login
!
line aux 0
!
line vty 0 4
 login local
 transport input ssh
!
!
!
End
```

Show IPv6 Route

IPv6 Routing Table - 16 entries

Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP

U - Per-user Static route, M - MIPv6

I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2

D - EIGRP, EX - EIGRP external

R 2003:BEBA:506:6::/64 [120/1]

via FE80::108:1, Serial0/1/0

R 2003:BEBA:506:7::/64 [120/1]

via FE80::108:1, Serial0/1/0

R 2003:BEBA:506:8::/64 [120/3]

via FE80::108:1, Serial0/1/0

R 2003:BEBA:506:9::/64 [120/3]

via FE80::108:1, Serial0/1/0

C 2003:BEBA:506:10::/64 [0/0]

via GigabitEthernet0/0, directly connected

L 2003:BEBA:506:10::1/128 [0/0]

via GigabitEthernet0/0, receive

C 2003:BEBA:506:11::/64 [0/0]

via GigabitEthernet0/1, directly connected

L 2003:BEBA:506:11::1/128 [0/0]

via GigabitEthernet0/1, receive

R 2003:BEBA:506:12::/64 [120/3]

via FE80::108:1, Serial0/1/0

R 2003:BEBA:506:13::/64 [120/3]

via FE80::108:1, Serial0/1/0

R 2003:BEBA:506:105::/64 [120/2]

via FE80::108:1, Serial0/1/0

R 2003:BEBA:506:106::/64 [120/2]

via FE80::108:1, Serial0/1/0

R 2003:BEBA:506:107::/64 [120/2]

via FE80::108:1, Serial0/1/0

C 2003:BEBA:506:108::/64 [0/0]

via Serial0/1/0, directly connected

L 2003:BEBA:506:108::2/128 [0/0]

via Serial0/1/0, receive

L FF00::/8 [0/0]

via Null0, receive

Show IPv6 interface brief

```
GigabitEthernet0/0          [up/up]
    FE80::10:1
    2003:BEBA:506:10::1
GigabitEthernet0/1          [up/up]
    FE80::11:1
    2003:BEBA:506:11::1
GigabitEthernet0/2          [administratively down/down]
    unassigned
Serial0/0/0                 [administratively down/down]
    unassigned
Serial0/0/1                 [administratively down/down]
    unassigned
Serial0/1/0                 [up/up]
    FE80::108:2
    2003:BEBA:506:108::2
Serial0/1/1                 [administratively down/down]
    unassigned
Serial0/2/0                 [administratively down/down]
    unassigned
Serial0/2/1                 [administratively down/down]
    unassigned
Serial0/3/0                 [administratively down/down]
    unassigned
Serial0/3/1                 [administratively down/down]
    unassigned
Vlan1                       [administratively down/down]
    unassigned
```

SUCURSAL 4

Show Running-config

```
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname SUCURSAL4
!
!
!
!
!
!
!
!
no ip cef
ipv6 unicast-routing
!
no ipv6 cef
!
!
!
username Manager secret 5 $1$mERr$9xcyzgMubYEUu5jrby8ds.
!
!
license udi pid CISCO2911/K9 sn FTX152479GH-
!
!
!
!
!
```

```
!
!
!
!
!
!
ip ssh version 2
no ip domain-lookup
ip domain-name red.local
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0
 description LAN 14
 no ip address
 duplex auto
 speed auto
 ipv6 address FE80::14:1 link-local
 ipv6 address 2003:BEBA:506:14::1/64
 ipv6 nd other-config-flag
 ipv6 nd managed-config-flag
 ipv6 rip ITLA_RIP enable
 ipv6 dhcp server LAN_14
!
interface GigabitEthernet0/1
```

```
 description LAN 15
 no ip address
 duplex auto
 speed auto
 ipv6 address FE80::15:1 link-local
 ipv6 address 2003:BEBA:506:15::1/64
 ipv6 nd other-config-flag
 ipv6 nd managed-config-flag
 ipv6 rip ITLA_RIP enable
 ipv6 dhcp server LAN_15
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Serial0/0/0
 description Conexion hacia ESTE
 no ip address
 ipv6 address FE80::101:1 link-local
 ipv6 address 2003:BEBA:506:101::1/64
 ipv6 rip ITLA_RIP enable
 clock rate 2000000
!
interface Serial0/0/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/1/0
```

```
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/1/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/2/0
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/2/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/3/0
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/3/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Vlan1
 no ip address
```

```
 shutdown
!
ipv6 router rip ITLA_RIP
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
 exec-timeout 5 0
 password 7 0822455D0A165546
 login
!
line aux 0
!
line vty 0 4
 login local
 transport input ssh
!
!
!
end
```

Show IPv6 Route

IPv6 Routing Table - 16 entries

```
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
R   2003:BEBA:506:1::/64 [120/1]
     via FE80::101:2, Serial0/0/0
R   2003:BEBA:506:2::/64 [120/1]
     via FE80::101:2, Serial0/0/0
R   2003:BEBA:506:3::/64 [120/3]
     via FE80::101:2, Serial0/0/0
R   2003:BEBA:506:4::/64 [120/2]
     via FE80::101:2, Serial0/0/0
R   2003:BEBA:506:5::/64 [120/4]
     via FE80::101:2, Serial0/0/0
C   2003:BEBA:506:14::/64 [0/0]
     via GigabitEthernet0/0, directly connected
L   2003:BEBA:506:14::1/128 [0/0]
     via GigabitEthernet0/0, receive
C   2003:BEBA:506:15::/64 [0/0]
     via GigabitEthernet0/1, directly connected
L   2003:BEBA:506:15::1/128 [0/0]
     via GigabitEthernet0/1, receive
C   2003:BEBA:506:101::/64 [0/0]
     via Serial0/0/0, directly connected
L   2003:BEBA:506:101::1/128 [0/0]
```

via Serial0/0/0, receive

R 2003:BEBA:506:102::/64 [120/2]

via FE80::101:2, Serial0/0/0

R 2003:BEBA:506:103::/64 [120/2]

via FE80::101:2, Serial0/0/0

R 2003:BEBA:506:104::/64 [120/3]

via FE80::101:2, Serial0/0/0

R 2003:BEBA:506:105::/64 [120/3]

via FE80::101:2, Serial0/0/0

L FF00::/8 [0/0]

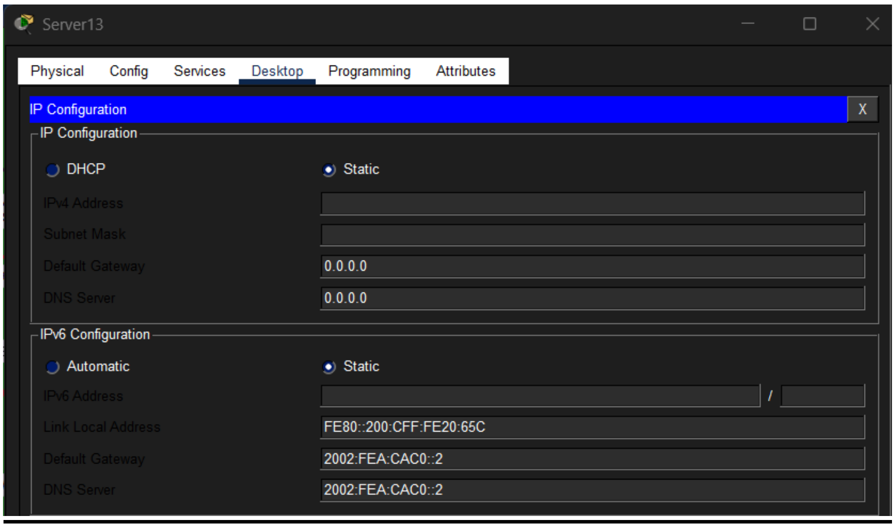
via Null0, receive

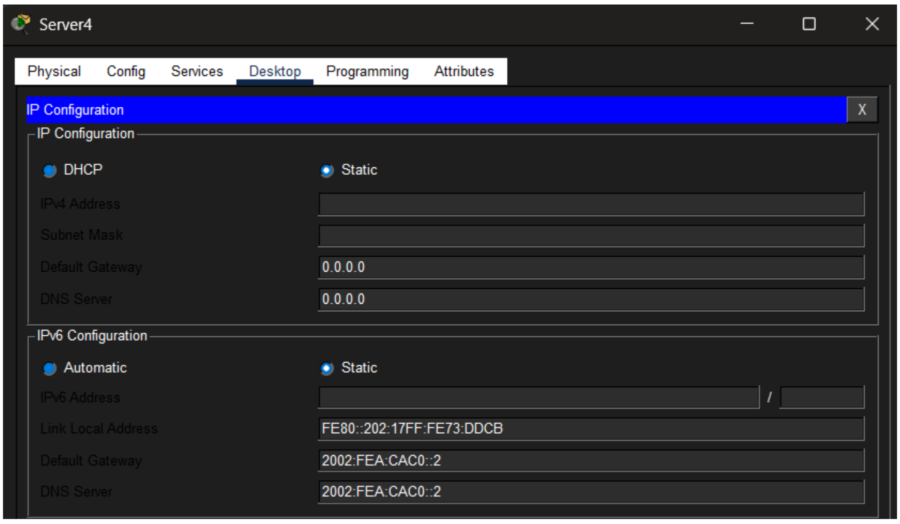
Show IPv6 interface brief

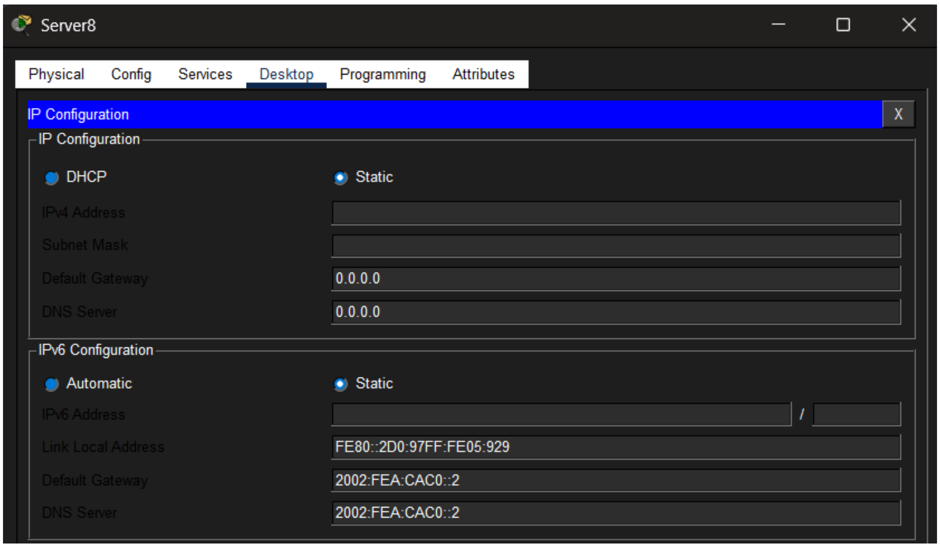
```
GigabitEthernet0/0          [up/up]
    FE80::14:1
    2003:BEBA:506:14::1
GigabitEthernet0/1          [up/up]
    FE80::15:1
    2003:BEBA:506:15::1
GigabitEthernet0/2          [administratively down/down]
    unassigned
Serial0/0/0                 [up/up]
    FE80::101:1
    2003:BEBA:506:101::1
Serial0/0/1                 [administratively down/down]
    unassigned
Serial0/1/0                 [administratively down/down]
    unassigned
Serial0/1/1                 [administratively down/down]
    unassigned
Serial0/2/0                 [administratively down/down]
    unassigned
Serial0/2/1                 [administratively down/down]
    unassigned
Serial0/3/0                 [administratively down/down]
    unassigned
Serial0/3/1                 [administratively down/down]
    unassigned
Vlan1                       [administratively down/down]
    unassigned
```

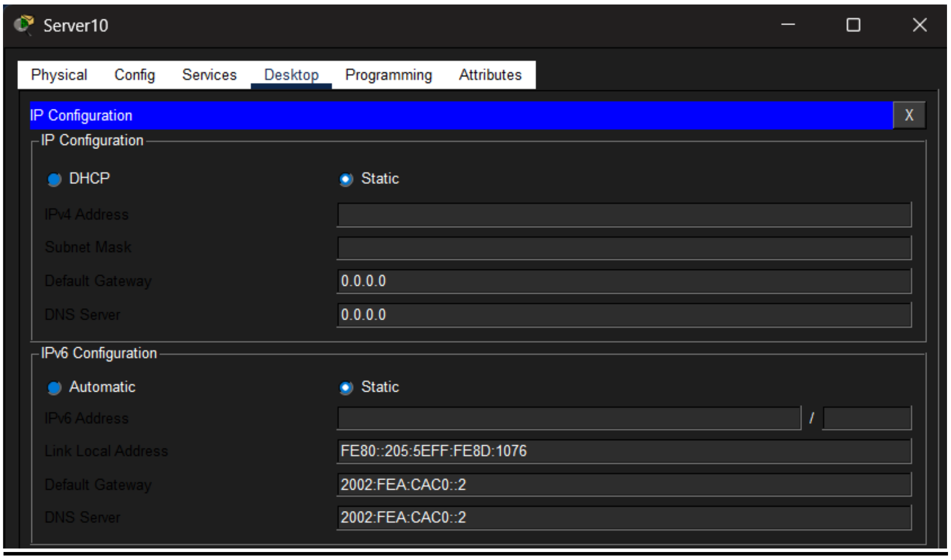
5.4 Evidencias (5 servidores DHCP y 5 hosts)

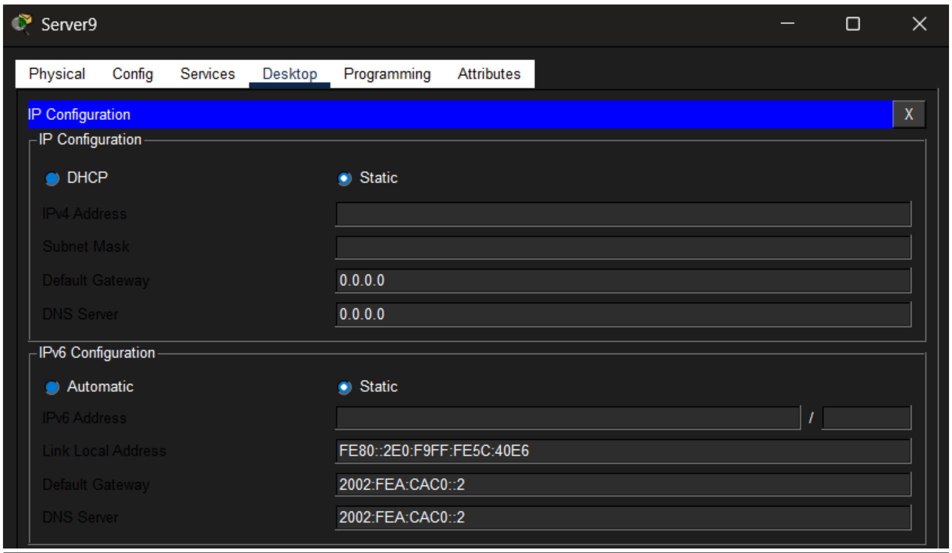
Servidores


Server13
Physical
Config
Services
Desktop
Programming
Attributes
IP Configuration
X
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::200:CFF:FE20:65C
Default Gateway
2002:FEA:CAC0::2
DNS Server
2002:FEA:CAC0::2

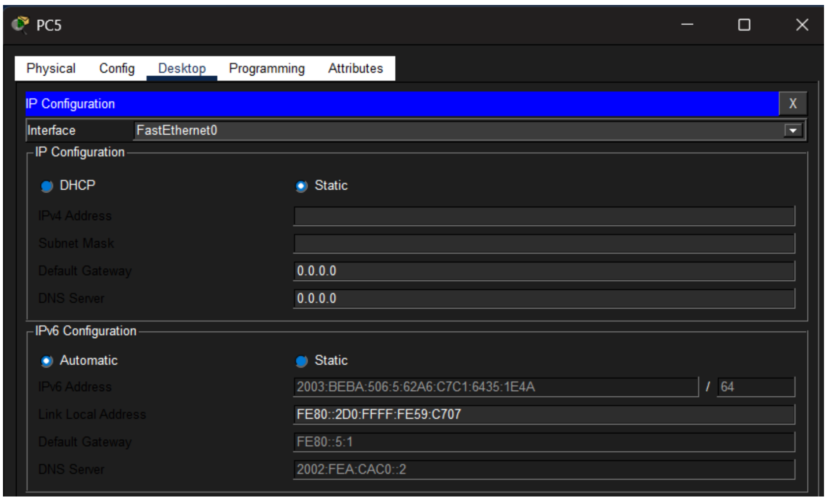

Server4
Physical
Config
Services
Desktop
Programming
Attributes
IP Configuration
X
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::202:17FF:FE73:DDCB
Default Gateway
2002:FEA:CAC0::2
DNS Server
2002:FEA:CAC0::2

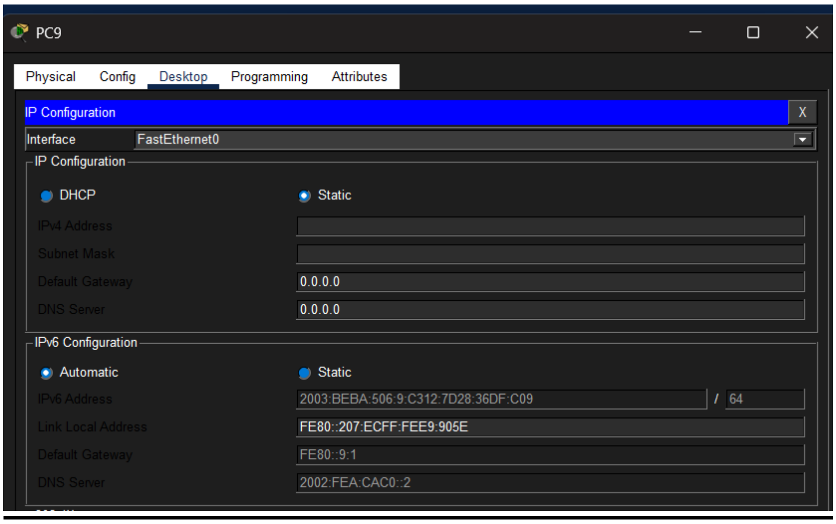

Server8
Physical
Config
Services
Desktop
Programming
Attributes
IP Configuration
X
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::2D0:97FF:FE05:929
Default Gateway
2002:FEA:CAC0::2
DNS Server
2002:FEA:CAC0::2

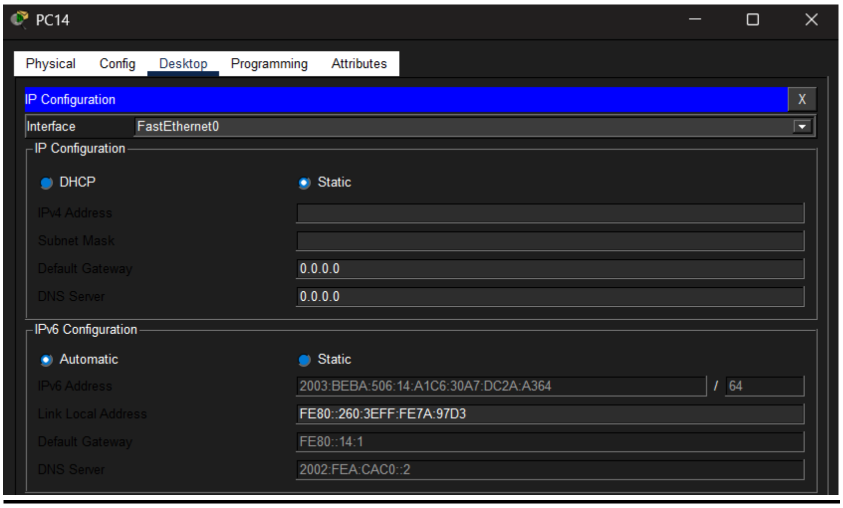

Server10
Physical
Config
Services
Desktop
Programming
Attributes
IP Configuration
X
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::205:5EFF:FE8D:1076
Default Gateway
2002:FEA:CAC0::2
DNS Server
2002:FEA:CAC0::2

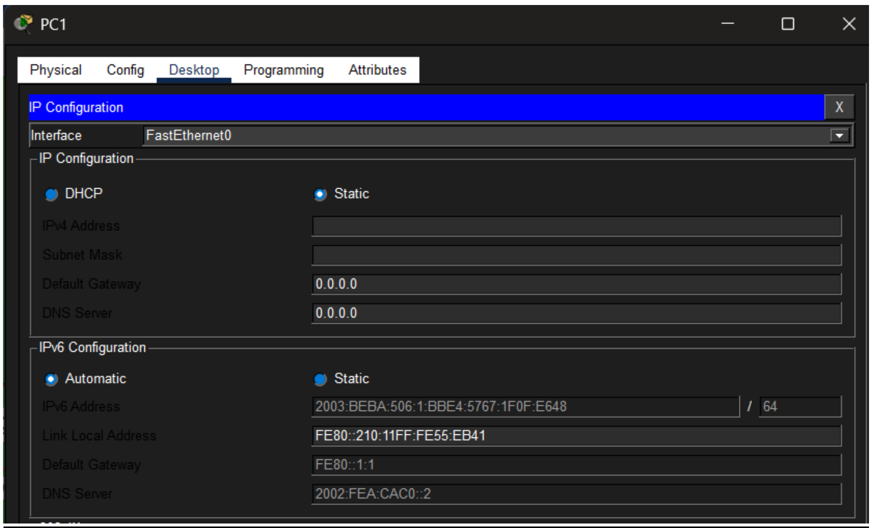

Server9
Physical
Config
Services
Desktop
Programming
Attributes
IP Configuration
X
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::2E0:F9FF:FE5C:40E6
Default Gateway
2002:FEA:CAC0::2
DNS Server
2002:FEA:CAC0::2

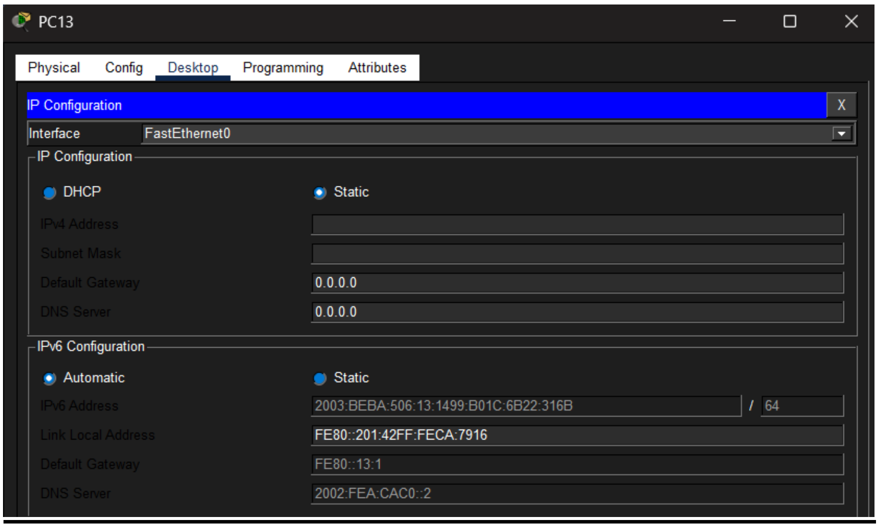
Hosts


PC5
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
2003:BEBA:506:5:62A6:C7C1:6435:1E4A
/
64
Link Local Address
FE80::2D0:FFFF:FE59:C707
Default Gateway
FE80::5:1
DNS Server
2002:FEA:CAC0::2


PC9
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
2003:BEBA:506:9:C312:7D28:36DF:C09
/
64
Link Local Address
FE80::207:ECFF:FEE9:905E
Default Gateway
FE80::9:1
DNS Server
2002:FEA:CAC0::2


PC14
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
2003:BEBA:506:14:A1C6:30A7:DC2A:A364
/
64
Link Local Address
FE80::260:3EFF:FE7A:97D3
Default Gateway
FE80::14:1
DNS Server
2002:FEA:CAC0::2


PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
2003:BEBA:506:1:BBE4:5767:1F0F:E648
/
64
Link Local Address
FE80::210:11FF:FE55:EB41
Default Gateway
FE80::1:1
DNS Server
2002:FEA:CAC0::2



5.5 Evidencias de conectividad (6 equipos remotos con ping exitoso)

	Exitoso	PC14	PC13	ICMPv6		0.000	N	0	(edit)
	Exitoso	PC16	PC7	ICMPv6		0.000	N	1	(edit)
	Exitoso	PC15	PC12	ICMPv6		0.000	N	2	(edit)
	Exitoso	PC16	PC10	ICMPv6		0.000	N	3	(edit)
	Exitoso	PC1	PC14	ICMPv6		0.000	N	4	(edit)
	Exitoso	PC0	PC4	ICMPv6		0.000	N	5	(edit)

	Exitoso	Sucursal_4	ESTE	ICMPv6		0.000	N	0	(edit)
	Exitoso	ESTE	NORESTE	ICMPv6		0.000	N	1	(edit)
	Exitoso	NORESTE	SEDE	ICMPv6		0.000	N	2	(edit)
	Exitoso	SUCURS...	SUCURSAL_1	ICMPv6		0.000	N	3	(edit)
	Exitoso	SUR	NORESTE	ICMPv6		0.000	N	4	(edit)
	Exitoso	SEDE	Sucursal_4	ICMPv6		0.000	N	5	(edit)

6. Comandos y Técnicas de Prueba y Solución de Problemas

Para garantizar el correcto funcionamiento de la topología IPv6 y resolver posibles fallos de comunicación, se utilizaron los siguientes comandos de diagnóstico en el CLI de Cisco:

Estos comandos permiten confirmar que las interfaces están activas y que las direcciones (Global Unicast y Link-Local) se asignaron correctamente.

Comando	**Propósito**
show ipv6 interface brief	Muestra un resumen del estado de todas las interfaces, sus direcciones IPv6 y si están en estado "up/up".
show ipv6 interface [interfaz]	Proporciona detalles específicos de una interfaz, incluyendo los grupos multicast a los que pertenece y el estado de los mensajes ICMPv6.
show running-config	Verifica que las configuraciones de SSH, contraseñas y enrutamiento RIPng se hayan aplicado según lo planeado.

Se utilizan para validar que el tráfico puede viajar de un extremo a otro de la red.

- **ping ipv6 [dirección_destino]**: Es la herramienta principal para verificar la conectividad de Capa 3. Un resultado exitoso indica que el host de destino es alcanzable.
- **traceroute ipv6 [dirección_destino]**: Permite identificar cada salto (router) que atraviesa el paquete. Es vital para localizar en qué punto de la red se está perdiendo la conexión.

Específicamente para este proyecto, donde se utiliza RIPng y asignación dinámica, los siguientes comandos son cruciales:

- **show ipv6 route**: Muestra la tabla de enrutamiento completa. Aquí verificamos que aparezcan las rutas aprendidas mediante el protocolo **RIP (marcadas con una 'R')**.
- **show ipv6 dhcp pool**: Permite confirmar que los prefijos y la información de DNS se están entregando correctamente a los dispositivos finales.
- **show ipv6 dhcp binding**: Muestra las direcciones IPv6 que han sido asignadas dinámicamente a los clientes (hosts).

Técnicas de Solución de Problemas (Troubleshooting)

1. **Verificación de Capa Física:** Confirmar que las interfaces (especialmente las Seriales, como la **Serial 0/2/0**) tengan el cableado correcto y la señal de reloj (*clock rate*) si es necesario.
2. **Validación de Link-Local:** Asegurarse de que las interfaces en el mismo segmento pueden verse usando sus direcciones fe80::, que son la base para la formación de adyacencias en RIPng.
3. **Revisión del Proceso RIPng:** Si una ruta no aparece en la tabla, se debe verificar que el comando ipv6 router rip [nombre] esté activo y que cada interfaz participante tenga el comando ipv6 rip [nombre] enable.

7. Organización en tabla

Categoría	**Comando de Cisco IOS**	**Función y Objetivo de la Técnica**
Conectividad Básica	**ping ipv6 [dirección]**	**Envía mensajes ICMPv6 Echo Request para verificar si un host remoto está activo y alcanzable.**
Rastreo de Ruta	**traceroute ipv6 [dirección]**	**Identifica cada salto (hop) en el trayecto hacia el destino, permitiendo localizar fallos en enlaces específicos.**
Interfaz y Capa 1/2	**show ipv6 interface brief**	**Verifica el estado físico (Up/Down) de las interfaces y confirma la asignación de las direcciones GUA y Link-Local.**
Enrutamiento	**show ipv6 route**	**Muestra la tabla de enrutamiento IPv6 para validar que las rutas RIP (R) y las redes directamente conectadas están presentes.**
Configuración	**show running-config**	**Permite revisar que los parámetros de SSH, el nombre del dominio (ITLA) y las contraseñas coincidan con el diseño.**
Gestión de IP	**show ipv6 dhcp pool**	**Valida que el servidor DHCPv6 esté configurado con el prefijo correcto y entregando parámetros de red.**
Acceso Remoto	**show ip ssh**	**Confirma que la versión 2 de SSH está habilitada y lista para recibir conexiones administrativas seguras.**

8. Buenas practicas de seguridad

Categoría	**Práctica de Seguridad**	**Descripción y Objetivo**	**Configuración Aplicada**
Acceso Administrativo	**SSH Versión 2**	**Sustituye a Telnet para cifrar todo el tráfico de administración remota, evitando la interceptación de credenciales.**	**ip ssh version 2**
Protección de Datos	**Cifrado de Contraseñas**	**Aplica un algoritmo de cifrado a todas las contraseñas en texto plano dentro del archivo de configuración.**	**service password-encryption**
Control de Acceso	**Autenticación Local**	**Exige un nombre de usuario y contraseña específicos (Base de Datos Local) para acceder a las líneas VTY.**	**login local**
Seguridad de Línea	**Restricción de Protocolo**	**Limita el acceso a las líneas virtuales únicamente al protocolo seguro SSH, bloqueando métodos inseguros.**	**transport input ssh**
Gestión de Sesión	**Timeout de Ejecución**	**Desconecta automáticamente las sesiones inactivas tras un tiempo determinado para evitar accesos no autorizados en terminales desatendidas.**	**exec-timeout 5 0**
Seguridad de Capa 2	**Deshabilitación de Puertos**	**Las interfaces que no están en uso se mantienen administrativamente apagadas para prevenir conexiones físicas no autorizadas.**	**shutdown**

Categoría	Práctica de Seguridad	Descripción y Objetivo	Configuración Aplicada
Infraestructura	Desactivación de Búsqueda DNS	Evita que el router intente traducir comandos mal escritos como nombres de dominio, lo que previene retrasos y posibles ataques de denegación de servicio local.	no ip domain-lookup
Identidad de Red	Dominio Específico	Define un contexto de seguridad y pertenencia organizacional para la generación de llaves criptográficas.	ip domain-name red.local

9. Conclusion

La infraestructura de red se basa en un modelo de administración segura donde el primer pilar es el control de acceso. Se ha implementado **SSH versión 2** en todos los dispositivos para garantizar que cualquier gestión remota viaje cifrada, eliminando vulnerabilidades asociadas a protocolos en texto plano. Complementando esto, el comando `service password-encryption` asegura que las credenciales locales no sean visibles en la configuración, mientras que el parámetro `login local` obliga a una validación estricta de usuarios. Para prevenir accesos accidentales o ataques por fuerza bruta en la consola, se utiliza `no ip domain-lookup` y se han configurado tiempos de espera de sesión mediante `exec-timeout`.

En cuanto a la conectividad y el mantenimiento, la visibilidad de la red se apoya en el uso de direcciones **Link-Local** personalizadas. Estas direcciones, configuradas manualmente con el formato `FE80::X:1`, permiten que el protocolo **RIPng** establezca adyacencias de vecinos de manera predecible y facilitan el diagnóstico rápido mediante `ping ipv6` y `traceroute ipv6`. El estado operativo de las interfaces se monitorea constantemente con `show ipv6 interface brief`, lo que permite confirmar que tanto las direcciones globales (GUA) como las físicas están en estado activo.

Finalmente, la gestión dinámica de los hosts se apoya en el servicio **DHCPv6 Stateful**. Los routers actúan como servidores asignando prefijos específicos a través de pools configurados, validando la entrega de direcciones con el comando `show ipv6 dhcp binding`. Para asegurar que los dispositivos finales reciban toda la información de red necesaria, las interfaces LAN incluyen los flags de configuración administrada, lo que centraliza el control del direccionamiento en toda la topología.